

Andrew J. Peng

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EDUCATION

Bachelor of Science in Mechanical Engineering

University of Maryland

Aug. 2024 – May 2028

College Park, MD

- **GPA:** 3.96 | A. James Clark School of Engineering Academic Honors
- **Coursework:** Linear Algebra, Differential Equations, Thermodynamics, Dynamics, Mechanics of Materials, Statics

EXPERIENCE

Game Developer

Contractor

Dec. 2024 – Present

Belmont, CA

- Utilized **Luau** to design and deliver **30+ modular game systems**, including data management, UI/UX, gameplay mechanics, and physics simulations for client commissions; **contributing to 15,000+ cumulative visits**.
- Led gameplay development for Tiny Lab Studio; authored a backend data service with **session-locking** and **JSON serialization**, supporting a **4,000+ member community**.

Mechanical Design Engineer

Terps Racing

Aug. 2024 – May 2026

College Park, MD

- Designed **ABS housings** using **SolidWorks** for FSAE EV brake temperature sensor and push rod strain gauge integration; performed FEA on sensor mounts to validate structural integrity against vibrational racing loads.
- Modeled a **clutch actuation** assembly using **SolidWorks** for the FSAE IC vehicle; performed FEA on the Onyx carbon lever and 6061 aluminum mount to validate structural integrity under **150N peak driver input** and vibrational loads.
- Contributed to vehicle manufacturing via **waterjet machining**, **3D printing**, and gusset installation, maintaining dimensional tolerances for chassis assembly.

PROJECTS

ATLAS: Autonomous Target Location Acquisition System

Apr. 2026 – May 2026

- Developed a computer vision pipeline using **YOLOv8** for inference, **PyTorch** for training, and **OpenCV** for screen capture and webcam feed, enabling live target and obstacle detection.
- Built a **semi-automated data pipeline** using a pretrained YOLOv8 model to generate annotation suggestions; manually reviewed and assembled **~18,000 frames** with an 80/20 train/val split, achieving **0.7+ confidence** on the custom dataset.
- Configured full **AdamW** training with cosine LR scheduling, warmup, and augmentation tuning; automated **ONNX/TensorRT export** for GPU-accelerated deployment.
- Implemented a **custom keyboard-driven annotation tool** for frame review, bounding box correction, and multi-class labeling with **YAML-based configuration**.

Autonomous Over-Terrain Vehicle

Aug. 2024 – Dec. 2024

- Engineered a two-wheel DC motor drivetrain using **Fusion 360**; wired motors to an **Arduino Uno** and authored a sequential navigation algorithm for obstacle avoidance using **HC-SR04** sensors.
- Integrated **ESP8266 Wi-Fi** telemetry to receive overhead camera positioning data via QR code tracking, achieving **sub 3-degree turn tolerance** for precise autonomous navigation.

SKILLS

Languages: Python, Luau, Java, MATLAB, HTML/CSS

Frameworks: PyTorch, OpenCV, YOLOv8, ONNX/TensorRT

Software: Git, SolidWorks, Fusion 360

Hardware: Arduino, Raspberry Pi